

4.5 Solution: (a) The force, by definition, is

$$\mathbf{F} = \int \rho(\mathbf{x}) \mathbf{E}^{(0)}(\mathbf{x}) d^3x.$$

Expand the electric field as

$$\mathbf{E}^{(0)}(\mathbf{x}) = \mathbf{E}^{(0)}(0) + \sum_i x_i \left. \frac{\partial \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i} \right|_{\mathbf{x}=0} + \frac{1}{2} \sum_{i,j} x_i x_j \left. \frac{\partial^2 \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i \partial x_j} \right|_{\mathbf{x}=0}.$$

Since

$$\nabla^2 \mathbf{E}^{(0)}(\mathbf{x}) = -\nabla^2(\nabla \Phi^{(0)}(\mathbf{x})) = -\nabla(\nabla^2 \Phi^{(0)}(\mathbf{x})) = 0,$$

we can add an extra

$$-\frac{1}{6} r^2 \nabla^2 \mathbf{E}^{(0)}(\mathbf{x}) \Big|_{\mathbf{x}=0} = -\frac{1}{6} r^2 \sum_{i,j} \delta_{ij} \left. \frac{\partial^2 \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i \partial x_j} \right|_{\mathbf{x}=0}$$

to the expansion of the electric field. Then,

$$\mathbf{E}^{(0)}(\mathbf{x}) = \mathbf{E}^{(0)}(0) + \sum_i x_i \left. \frac{\partial \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i} \right|_{\mathbf{x}=0} + \frac{1}{6} \sum_{i,j} (3x_i x_j - r^2 \delta_{ij}) \left. \frac{\partial^2 \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i \partial x_j} \right|_{\mathbf{x}=0}.$$

Upon integration with the charge density, we will have

$$\mathbf{F} = q \mathbf{E}^{(0)}(0) + \sum_i p_i \left. \frac{\partial \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i} \right|_{\mathbf{x}=0} + \frac{1}{6} \sum_{i,j} Q_{ij} \left. \frac{\partial^2 \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i \partial x_j} \right|_{\mathbf{x}=0},$$

where q , $\mathbf{p}$, and Q_{ij} are as defined in the text. Using the relation

$$\nabla(\mathbf{a} \cdot \mathbf{b}) = (\mathbf{a} \cdot \nabla) \mathbf{b} + (\mathbf{b} \cdot \nabla) \mathbf{a} + \mathbf{a} \times (\nabla \times \mathbf{b}) + \mathbf{b} \times (\nabla \times \mathbf{a}),$$

and notice that $\mathbf{p}$ and Q_{ij} are pure numbers and $\nabla \times \mathbf{E} = 0$, we can write the force as

$$\mathbf{F} = q \mathbf{E}^{(0)}(0) + \left\{ \nabla[\mathbf{p} \cdot \mathbf{E}^{(0)}(\mathbf{x})] \right\}_0 + \left\{ \nabla \left[\frac{1}{6} \sum_{i,j} Q_{ij} \frac{\partial E_i^{(0)}(\mathbf{x})}{\partial x_j} \right] \right\}_0.$$

(b) The torque is defined as

$$\begin{aligned} \mathbf{N} &= \int \mathbf{x} \times [\rho(\mathbf{x}) \mathbf{E}^{(0)}(\mathbf{x})] d^3x = \int [\rho(\mathbf{x}) \mathbf{x}] \times \mathbf{E}^{(0)}(\mathbf{x}) d^3x \\ &= \int [\rho(\mathbf{x}) \mathbf{x}] \times \mathbf{E}^{(0)}(0) d^3x + \int [\rho(\mathbf{x}) \mathbf{x}] \times \sum_i x_i \left. \frac{\partial \mathbf{E}^{(0)}(\mathbf{x})}{\partial x_i} \right|_{\mathbf{x}=0} d^3x. \end{aligned}$$

The first term is clearly

$$\mathbf{p} \times \mathbf{E}^{(0)}(0).$$

For the second term, consider its i -th component, we have

$$\epsilon_{ijk} \sum_l \int \rho(\mathbf{x}) x_j x_l \left. \frac{\partial E_k^{(0)}(\mathbf{x})}{\partial x_l} \right|_{\mathbf{x}=0} d^3x = \frac{1}{3} \epsilon_{ijk} \sum_l Q_{jl} \left. \frac{\partial E_k^{(0)}(\mathbf{x})}{\partial x_l} \right|_{\mathbf{x}=0},$$

where we have included a term that evaluates to 0. Again, $\nabla \times \mathbf{E} = 0$ implies that

$$\frac{\partial E_k^{(0)}(\mathbf{x})}{\partial x_l} = \frac{\partial E_l^{(0)}(\mathbf{x})}{\partial x_k}.$$

Then, the second term becomes

$$\frac{1}{3}\epsilon_{ijk} \sum_l Q_{jl} \frac{\partial E_l^{(0)}(\mathbf{x})}{\partial x_k} \Big|_{\mathbf{x}=0} = \frac{1}{3}\epsilon_{ijk} \frac{\partial}{\partial x_k} \left[\sum_l Q_{jl} E_l^{(0)}(\mathbf{x}) \right] \Big|_{\mathbf{x}=0},$$

which agrees with the second term in the problem for $i = 1$.